Logan Robinson

PhD Candidate · Mechanical Engineering Background · Logan@gui.us · San Antonio, TX · LinkedIn

Summary

Mechanical engineer and PhD candidate with experience spanning nuclear systems, advanced materials, experimental hardware, computational methods, and scientific data analysis. Experience includes reactor experiment hardware design at Idaho National Laboratory, nuclear process data analysis at Pacific Northwest National Laboratory, and materials characterization research for advanced manufacturing. Combines mechanical engineering fundamentals with computational and data-driven methods to analyze complex engineering systems, interpret experimental data, and develop analytical and predictive tools.

Publications

- **Robinson, L.** et al. "Beyond Energy Density: Multimodal Learning for Porosity Prediction in LPBF." SPIE Defense + Security, 2026.
- **Robinson, L.** et al. "Evaluation and a New Benchmark for Defect Detection in Materials Fabrication." CVPR MM4MAT Workshop, 2025.
- **Robinson, L.** et al. "Multimodal Approaches for Realistic Synthetic Data Generation in Materials Fabrication." CVPR MM4MAT Workshop, 2025.

Education

University of Texas at San Antonio — San Antonio, TX
PhD Computer Science (Coursework Complete – GPA: 3.91) — *Estimated: Dec 2028*

University of Texas at San Antonio — San Antonio, TX
BS Mechanical Engineering — *Dec 2022*

Machine Learning Research and Engineering

Vision & Artificial Intelligence Lab, UTSA — San Antonio, TX
Graduate Researcher — *2023 – Present*

- Designed machine learning systems for industrial inspection and scientific imaging, including a novel multimodal fusion architecture for integrating image and process information.
- Uncovered a difficulty-stratified dissociation between detection and retrieval performance detection remains stable while retrieval collapses on hard samples a failure mode invisible to standard aggregate metrics
- Developed automated data curation and evaluation pipelines to improve model robustness and ensure rigorous performance assessment across benchmark datasets.
- Developed a few-shot segmentation model for laser powder bed fusion (LPBF) stainless steel defect detection, achieving 0.926 AUC on held-out defect classes; delivered production inference pipeline for lab handoff.

Pacific Northwest National Laboratory (PNNL) — Richland, WA
AI/ML Research Intern — *2024*

- Developed machine learning models to analyze heterogeneous process sensor data from advanced nuclear fuel recycling operations.
- Built data processing, feature engineering, and model evaluation pipelines supporting process monitoring and anomaly detection for complex nuclear process data.
- Applied interpretable machine learning methods to relate model predictions to underlying process behavior, enabling validation with domain experts and improving confidence in model outputs.

Idaho National Laboratory — Idaho Falls, ID
Mechanical Design Intern — *2022 – 2023*

- Designed mechanical systems supporting reactor experiments, including a retractable in-core sensor deployment mechanism that reduced system footprint by approximately 40% through iterative engineering.

– Modeled reactor experiment hardware in Autodesk Inventor and fabricated rapid prototypes to evaluate fit, manufacturability, mechanical clearances, and thermal performance.

– Evaluated experimental hardware performance and incorporated testing results into subsequent design iterations to improve robustness in demanding operating environments.

Extreme Environments Materials Lab, UTSA Feb 2023 – Aug 2023
Research Fellow *San Antonio, TX*

– Led highly controlled, data-driven synthesis experiments, managing multi-variable processing conditions to generate reliable baseline data for downstream machine learning tasks.

– Executed advanced microstructural image analytics and maintained end-to-end reproducible data pipelines to support tier-one research output and future publishing.

UTSA College of Engineering San Antonio, TX
Laboratory Assistant *2019 – 2022*

– Trained over 100 students on the safe, repeatable operation of machine shop equipment (lathes, mills, CO_2 laser cutter) and authored standard operating procedures (SOPs).

– Fabricated precision prototype components using a mix of traditional machining and multi-material 3D printing techniques.

SwRI / UTSA Capstone San Antonio, TX
Design Engineer *2022*

– Designed a propellant management device (PMD) using surface tension geometry to control fluid transport in a tightly constrained environment.

– Delivered a manufacturable design within a 15% budget constraint, meticulously balancing micro-porosity boundaries and structural strength requirements.

– Validated structural sealing compliance and physical geometry limits against predictive CFD simulations and experimental test data.

Technical Skills

- **Engineering / CAD:** SOLIDWORKS, ANSYS, Autodesk Inventor, PTC Creo, CFD, GD&T
- **Fabrication & Test:** Machining (lathe, mill), Multi-material 3D Printing, Silicone Molding & Bonding, Prototyping, LabVIEW
- **Machine Learning:** PyTorch, Computer Vision, Transformers, Multimodal Fusion, Segmentation, Self-Supervised / Foundation Models (DINOv3)
- **Programming & Systems:** Python, C, SQL, MATLAB/Simulink, Git, Linux, Docker, HPC

Awards & Memberships

- **First Place, Tech Symposium** – The University of Texas at San Antonio, 2022
- **Programs Chair**, American Institute of Aeronautics and Astronautics (AIAA), 2021 – 2022
- **Treasurer**, Texas Society of Professional Engineers, 2022